

SMART DISASTER RELIEF INVENTORY AND DISTRIBUTION MANAGEMENT SYSTEM USING C

CSA0203 – C PROGRAMMING

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PROJECT SUMMARY

The Smart Disaster Relief Inventory and Distribution Management System is a menu-driven C application developed to efficiently manage essential relief resources during natural and man-made disasters such as floods, earthquakes, cyclones, landslides, forest fires and industrial accidents. The project addresses the problems associated with manual inventory management, including inaccurate stock records, duplicate entries, resource shortages, inefficient distribution and difficulty in identifying resources that require immediate replenishment. Each resource is maintained with details such as Resource ID, name, category, available quantity, minimum required quantity, distribution centre and priority. The system supports resource addition, updating, display, searching, sorting, allocation/distribution, analysis, merging of relief-centre records, file storage and report generation. Resources can be categorized into Food and Water, Medicine, Rescue Equipment, Transportation, Shelter, Clothing, Communication and Other resources. The analysis module compares available stock with minimum requirements and classifies resources as ADEQUATE, LOW or CRITICAL, enabling administrators to identify shortages and plan replenishment. During distribution, the system checks available stock and prevents allocation beyond the quantity in inventory. Records from another relief centre can be merged while avoiding duplicate Resource IDs. Inventory data is stored permanently using files, with resources.dat used for saving/loading records and disaster_report.txt used for generating resource and replenishment reports.

DESIGN AND IMPLEMENTATION

The application uses an array of structures to store multiple resource records and user-defined functions to keep the program modular and maintainable. Pointers provide efficient access to resource records, while recursion is used for resource analysis. Loops and decision-making statements support searching, validation, classification and inventory processing. Bubble Sort is used to organize resources by quantity, priority, category or distribution centre. The system is designed for C using GCC/MinGW/Clang and can be developed in VS Code, Code::Blocks or Dev-C++ on Windows, Linux or macOS.

KEY FUNCTIONALITIES

Module	Functionality
Resource Management	Add, update, delete and display relief-resource records.
Search & Sorting	Search by ID/name/category/centre; sort by quantity, priority, category or centre.
Analysis	Calculate total stock and classify resources as Adequate, Low or Critical.
Distribution	Validate available stock and distribute resources without exceeding inventory.
Multi-Centre Merge	Merge records from relief centres while preventing duplicate IDs.
File & Reporting	Save/load inventory and generate disaster-resource reports.

TESTING AND RESULT

The system was tested with valid and invalid operations including resource addition, duplicate-ID detection, searching, shortage analysis, valid and excess distribution, sorting, file saving, report generation and merging. All 18 documented test cases in the report produced PASS results, confirming the expected behaviour of the major modules.

OVERALL OUTCOME

The project demonstrates the practical application of operators, decision-making, switch-case, loops, arrays, strings, structures, functions, pointers, recursion, searching, sorting, merging and file handling to a real-world disaster-management problem. It provides an organized approach for monitoring stock, supporting timely resource distribution and identifying critical shortages. The console-based system can be extended in future with databases, cloud storage, GPS tracking, IoT sensors, mobile applications and artificial intelligence.